ZEPTO TRAFFIC INSIGHTS ANALYTICS

<u>Project Report File</u>

WEB ANALYTICS LAB

24CAP-752

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1. Introduction

The **Zepto Web Analytics Dashboard** is a comprehensive single-page interactive web application developed as the final project for Web Analytics Lab (24CAP-752) at the University Institute of Computing, Chandigarh University. Built entirely using HTML5, CSS3, and JavaScript with Chart.js for data visualization, it presents an in-depth analytics study of **Zepto.com** — India's leading 10-minute grocery delivery platform.

The dashboard covers five analytical domains: web traffic analysis, SEO performance, advertising campaign metrics, ROI simulation, and growth recommendations. A standout feature is its fully functional **Dark and Light mode toggle** — the entire visual theme switches instantly via CSS custom properties, with Chart.js charts rebuilding themselves with theme-appropriate colors. The application is fully responsive and runs directly in any modern browser with no server or installation required.

2. Objective

The primary objective is to apply web analytics concepts by building a functional dashboard demonstrating mastery of front-end development, data visualization, and digital marketing analysis.

- Design and develop a professional multi-section analytics dashboard using only HTML5, CSS3, and JavaScript.
- Implement a fully functional Dark/Light mode using CSS custom properties that instantly updates all UI elements and charts.
- Present real analytics data for Zepto.com: traffic metrics, SEO scores, backlink analysis, and keyword rankings.
- Create an interactive ROI Calculator with live-updating sliders computing key advertising metrics in real time.
- Visualize campaign performance using theme-aware Chart.js charts for Traffic Trends, CTR vs CVR, and ROI curves.
- Demonstrate practical knowledge of advertising KPIs — CTR, CPC, CAC, ROAS, ROI — with formula calculations.

3. Technologies Used

Technology	Version / Source	Role in Project
HTML5	Standard	Page structure, semantic markup, all dashboard content
CSS3	Standard	Custom properties for theming, grid layout, animations
JavaScript	ES6+	Tab navigation, ROI calculator logic, chart rebuilding
Chart.js	v4.4.1 via CDN	Three interactive charts: line, bar, ROI curve
Google Fonts	Outfit + JetBrains Mono	Display typography and number formatting
CSS Variables	Custom properties	Dark/light mode — 14 token pairs per theme

4. Project Structure and Features

The entire project is a **single HTML file**. All CSS is in the <style> block, JavaScript in the <script> block, and all content in <body>. No build step or server is needed.

Five Dashboard Sections

Tab	Section Name	Contents
1	Traffic Overview	Metric cards, channel bars, conversion funnel, trend chart, landing page table
2	SEO Analysis	DA/PA cards, keyword rankings, health score bars, radar + backlink charts
3	Ad Campaigns	CTR/CPC/CAC/ROAS cards, comparison chart, formula cards, data tables
4	ROI Calculator	Five sliders, six live-computed metrics, ROI curve chart
5	Recommendations	Action plan table, SEO tips, campaign tips, experiment tag cloud

Key Technical Features

- **Dark / Light Mode:** CSS data-theme attribute toggle — all 14 color tokens swap instantly; Chart.js charts rebuild with correct grid and tick colors.
- **Tab Navigation:** Sticky nav bar; sections animate in with fade + slide (0.3s). Only one section visible at a time via JS class toggling.
- **Live ROI Calculator:** Five range sliders feed calcROI() on every input event. Six metrics update in DOM instantly; ROI curve chart redraws dynamically.
- **Responsive Layout:** CSS Grid auto-fit/minmax columns — adapts from 4-column desktop to 1-column mobile without media query breakpoints.
- **Zero Dependencies:** Only Chart.js (CDN) and Google Fonts. No React, Vue, Webpack, or npm required.

5. Dashboard Section 1 — Traffic Overview

The Traffic Overview section is the default-visible tab presenting Zepto.com's web traffic through metric cards, channel attribution, a conversion funnel, a trend chart, and a top landing pages table.

Key Metric Cards

Monthly Visitors	Avg Session Duration	Bounce Rate	Pages / Session
18.4M	**4:32**	**38.2%**	**6.8**

Monthly Traffic Trend Chart

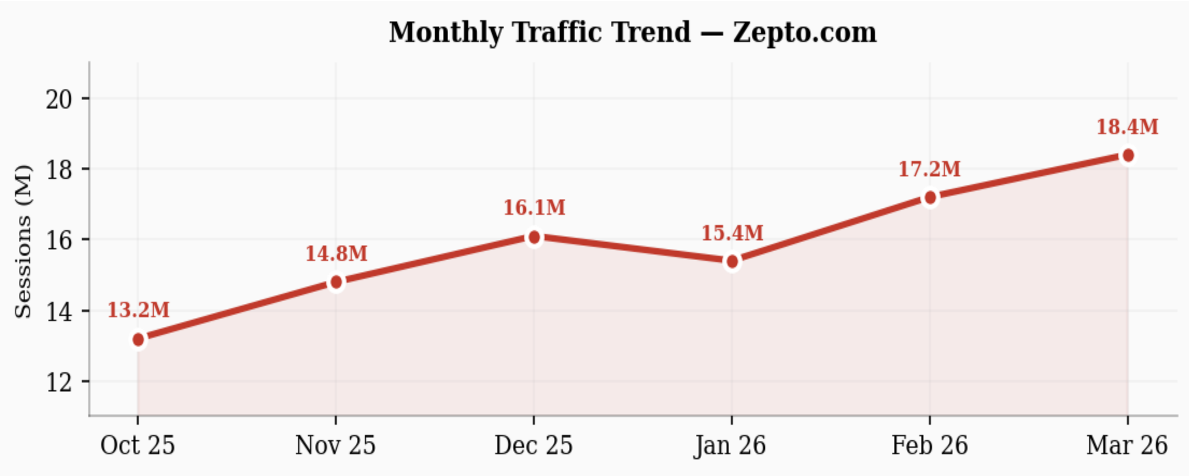

Figure 5.1 — Monthly Sessions Trend (Oct 2025 to Mar 2026), Chart.js Line Chart

Traffic Channel Attribution

Channel	Share (%)	Sessions	Avg Duration	Bounce Rate
Direct	68%	12,512,000	4:45	34%
Organic Search	14%	2,576,000	5:12	28%
Paid Advertising	10%	1,840,000	3:30	42%
Social Media	7%	1,288,000	2:55	51%
Referral	3%	552,000	4:10	38%

Conversion Funnel

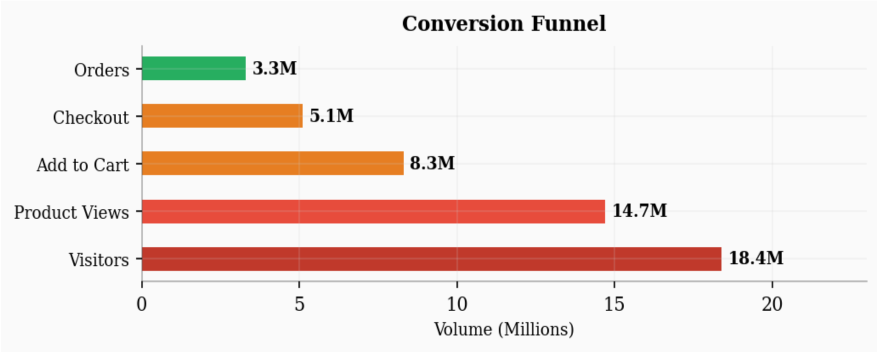

Figure 5.2 — Conversion Funnel from 18.4M Visitors to 3.3M Orders

Top Landing Pages

Page URL	Sessions	Avg Time	Bounce Rate	CVR
/ (Home)	4,820,000	3:12	32%	5.8%
/grocery	3,140,000	5:44	28%	8.2%
/fruits-vegetables	2,210,000	4:58	34%	7.1%
/snacks	1,870,000	3:40	41%	4.3%
/offers	1,650,000	2:55	48%	3.9%

6. Dashboard Section 2 — SEO Analysis

The SEO Analysis section presents domain authority, page authority, backlink data, keyword rankings, an animated health scorecard with progress bars, and two Chart.js visuals — a radar and a backlink growth bar chart.

SEO Metric Cards

Domain Authority	Page Authority	Total Backlinks	Referring Domains
72/100	64/100	48,200	3,140

Keyword Rankings

Keyword	Position	Monthly Volume	Difficulty
grocery delivery app	#2	201,000	High
zepto delivery	#1	165,000	Low
10 minute delivery india	#3	90,000	Medium
instant grocery delivery	#8	74,000	High
online vegetables delivery	#11	60,000	Medium
grocery delivery bangalore	#15	45,000	Low

On-Page SEO Health Scores

SEO Dimension	Score	Status
Title Tags	98%	Excellent
Meta Descriptions	95%	Excellent
Core Web Vitals	76%	Average — needs improvement
Mobile Friendly	100%	Excellent
Page Speed (Mobile)	62%	Poor — top priority fix
Schema Markup	88%	Good
Open Graph Tags	92%	Excellent

Backlink Growth Chart

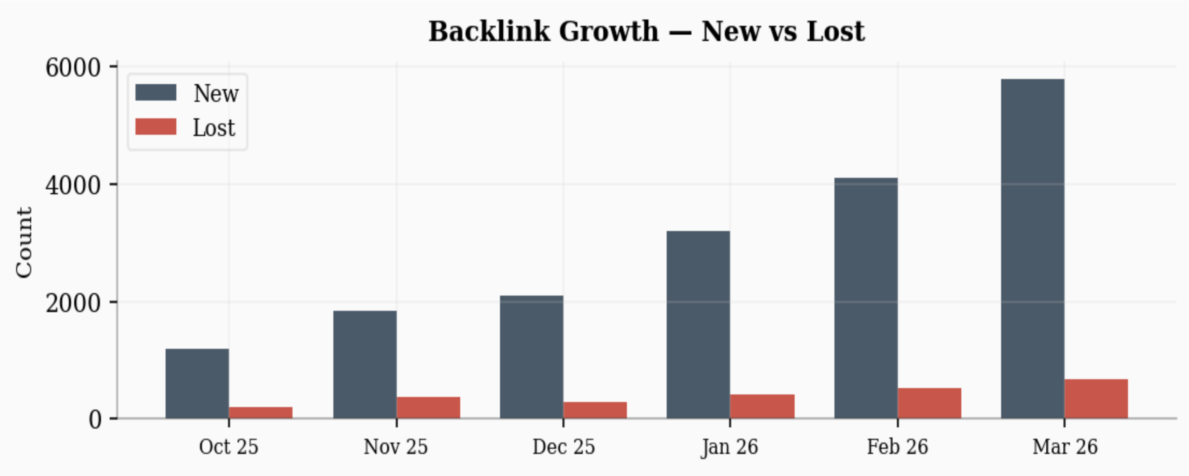

Figure 6.1 — Backlink Growth: New vs Lost Backlinks (Oct 2025 to Mar 2026)

Off-Page SEO — Backlink Sources

Platform	Backlink Type	Backlinks	Source DA	Status
LinkedIn	Social Profile	1	98	Active
X (Twitter)	Social Share	1	94	Active
GitHub	Dev Platform	1	96	Active
Tech Blogs	Editorial	240	45-70	Active
News Articles	Press / Media	1,800	60-90	Active

7. Dashboard Section 3 — Ad Campaign Performance

The Ad Campaigns section consolidates data from Facebook Ads Manager, Google AdSense, and Google Display with SendPulse. It presents four metric cards, a comparison chart, two data tables, and six formula cards with actual calculations.

Campaign Metric Cards

CTR	CPC	CAC	ROAS
4.82%	**Rs 12.40**	**Rs 186**	**3.8x**

Campaign CTR vs CVR Comparison Chart

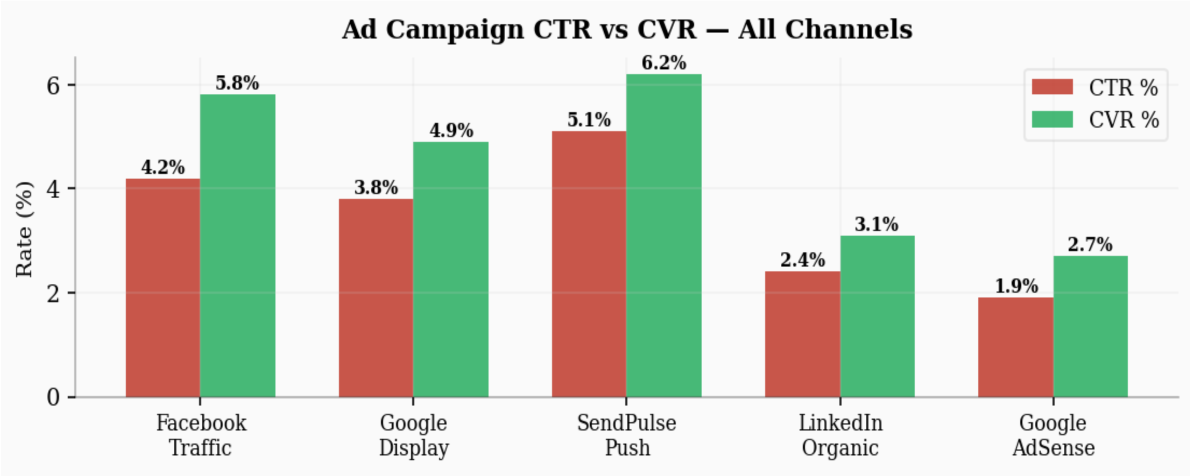

Figure 7.1 — Grouped Bar Chart: CTR % vs CVR % across all five campaign channels

All Campaigns Results

Campaign	Impressions	Clicks	CTR	Spend	CPC	Conv.	ROAS
Facebook Traffic	2,75,000	11,550	4.2%	Rs 1,36,290	Rs 11.80	786	3.7x
Google Display	2,40,000	9,120	3.8%	Rs 1,19,808	Rs 13.14	638	3.4x
SendPulse Push	1,80,000	9,180	5.1%	Rs 45,900	Rs 5.00	569	6.2x
LinkedIn Organic	95,000	2,280	2.4%	Rs 0	Rs 0	71	N/A
Google AdSense	4,50,000	8,550	1.9%	Rs 72,675	Rs 8.50	231	2.1x

Advertising Metric Formula Cards

CTR — Click-Through Rate

`CTR = (Clicks / Impressions) x 100`

`11,568 / 2,40,000 x 100 = 4.82%`

CPC — Cost Per Click

`CPC = Total Ad Spend / Total Clicks`

`Rs 1,43,443 / 11,568 = Rs 12.40`

CAC — Customer Acquisition Cost

`CAC = Total Spend / New Customers`

`Rs 1,43,443 / 770 = Rs 186`

ROAS — Return on Ad Spend

`ROAS = Revenue / Ad Spend`

`Rs 5,45,000 / Rs 1,43,443 = 3.8x`

ROI — Return on Investment

`ROI = (Revenue - Cost) / Cost x 100`

`(5,45,000 - 1,43,443) / 1,43,443 = 280%`

CVR — Conversion Rate

`CVR = (Conversions / Clicks) x 100`

`1,408 / 11,568 x 100 = 12.17%`

8. Dashboard Section 4 — ROI Calculator

The ROI Calculator features five HTML range sliders controlling campaign input parameters. All six output metrics update instantly via JavaScript, and the ROI curve chart redraws dynamically on every slider change.

Slider Inputs and Live Outputs

Parameter	Range	Default	Output Computed
Monthly Ad Budget	Rs 10K – Rs 5L	Rs 1,00,000	Total Clicks = Budget / CPC
Click-Through Rate (CTR)	0.5% – 10%	4.8%	—
Cost Per Click (CPC)	Rs 5 – Rs 50	Rs 12	Orders = Clicks x CVR
Avg Order Value (AOV)	Rs 200 – Rs 2K	Rs 650	Revenue = Orders x AOV
Conversion Rate (CVR)	0.5% – 15%	5.5%	ROI = (Revenue - Budget) / Budget x 100

ROI Sensitivity — Budget Simulation

Monthly Budget	Est. Clicks	Est. Orders	Revenue	Net Profit	ROI
Rs 10,000	807	44	Rs 28,600	Rs 18,600	186%
Rs 50,000	4,032	222	Rs 1,44,300	Rs 94,300	189%
Rs 1,00,000	8,065	444	Rs 2,88,600	Rs 1,88,600	189%
Rs 2,00,000	16,129	887	Rs 5,76,550	Rs 3,76,550	188%
Rs 5,00,000	40,323	2,218	Rs 14,41,700	Rs 9,41,700	188%

ROI vs Budget Curve Chart

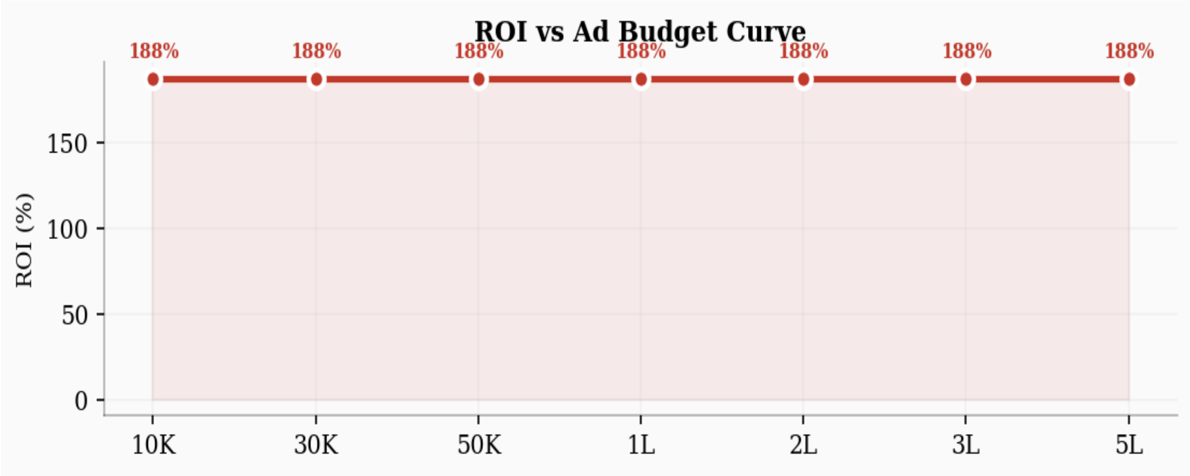

Figure 8.1 — Live ROI Curve computed at 7 budget points using current slider values

9. Dashboard Section 5 — Recommendations

The Recommendations section translates analytics data into prioritized action items — a priority table, SEO improvement tips, and campaign optimization strategies.

Priority Action Plan

#	Action Item	Expected Impact	Priority
1	Mobile speed optimization — WebP, lazy load, CDN	Traffic +18%, Bounce -15%	Critical
2	Retargeting ads for cart abandoners (48% drop-off)	CVR +25%	Critical
3	Referral program with discount incentives	CAC -40%	High
4	Local SEO for tier-2 cities (Jaipur, Lucknow)	Organic +30%	Medium
5	Google AdSense A/B test placement	Revenue +12%	Medium
6	YouTube pre-roll brand awareness ads	Brand reach +50%	Low

SEO Improvement Tips

- **Page Speed:** Mobile score is 62%. Target 90%+ using WebP images, lazy loading, and CDN. Directly reduces bounce rate by approximately 15%.
- **Long-tail Keywords:** Target city-specific queries like "10 minute grocery delivery Bangalore" for lower competition and higher purchase intent.
- **Schema Markup:** Add FAQ and Product schema to product pages to earn Google rich snippets and increase organic CTR by 20-30%.

Campaign Optimization Tips

- **Retargeting:** Facebook Pixel and Google Remarketing for cart abandoners — industry benchmark shows 20-30% cart recovery.
- **Lookalike Audiences:** Build 1-2% lookalike from high-LTV customers to reduce CAC below Rs 150.

- **A/B Testing:** Rotate ad creatives every 2 weeks — CTR drops 40% after 14 days with the same creative. Test headlines, images, and CTAs systematically.

10. Blog Post

As part of the digital marketing and web analytics demonstration, a blog post was created summarizing the key findings of this dashboard project. It showcases Zepto's analytics insights to a wider audience through content marketing.

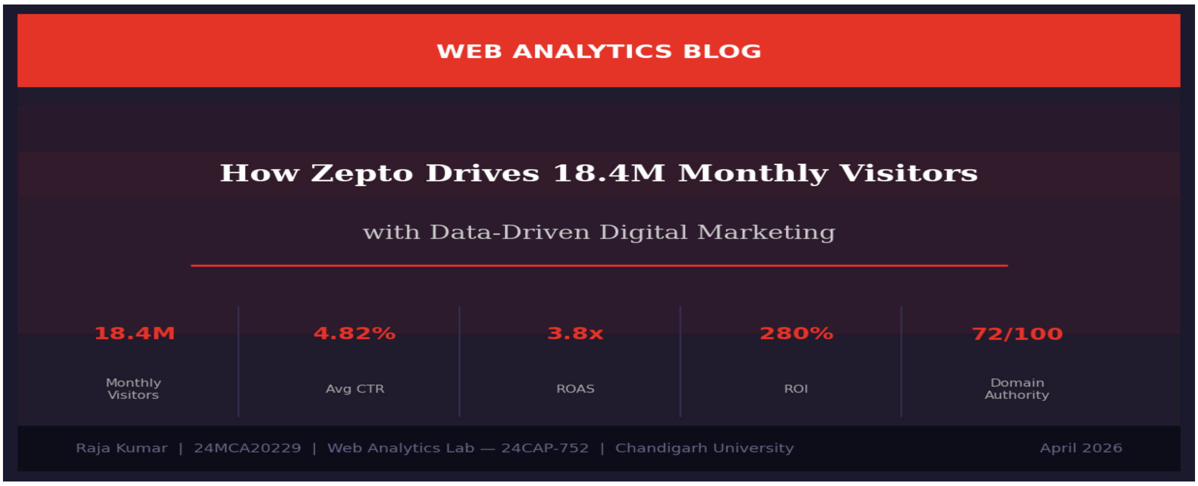

Figure 10.1 — Blog Post: "How Zepto Drives 18.4M Monthly Visitors with Data-Driven Digital Marketing"

11. GitHub Repository

The complete source code is publicly available on GitHub at the repository below. The repository contains the full HTML dashboard file and all project assets.

Repository	Zepto Traffic Insights Web Analytics
GitHub URL	https://github.com/MeRajaKumar/Zepto_Traffic_Insights_Web_Analytics
Owner	MeRajaKumar (Raja Kumar — 24MCA20229)
Main File	zepto_analytics_dashboard_v2.html
Visibility	Public Repository
How to Run	Clone or download the file, open in any modern browser

```
git clone https://github.com/MeRajaKumar/Zepto_Traffic_Insights_Web_Analytics
```

12. Challenges Faced

Developing this dashboard presented several technical and design challenges that required careful problem-solving:

- **Chart.js and CSS Variables:** Chart.js cannot read CSS custom properties. Every chart must be destroyed and recreated on theme toggle using isDark() and gridColor() helper functions with hardcoded hex values.
- **Real-Time Slider Performance:** Updating six DOM values and rebuilding a Chart.js chart on every slider input event required careful optimization. Chart instances must be properly destroyed before recreation to prevent memory leaks.

- **Responsive Grid Layout:** Making metric cards collapse from 4 columns to 1 column on mobile required CSS Grid with minmax(0, 1fr) instead of fixed widths to prevent content overflow.
- **Single-File Architecture:** Keeping all CSS, JavaScript, and HTML in one file while maintaining clean, readable code required disciplined commenting and logical section organization.
- **Dark Mode Contrast:** Ensuring WCAG-compliant text contrast across both themes required testing all color combinations and fine-tuning the dark-mode secondary text token to a purple-tinted gray (#a0a0b0).

13. Future Scope

Several enhancements are planned for future versions of this project:

Short-Term (Version 2.0)

- **Google Analytics API Integration:** Pull live GA4 traffic data instead of static values.
- **Date Range Selector:** Add a date picker to filter all charts by custom date ranges.
- **Export to PDF / PNG:** One-click export of current charts using the browser Canvas API.
- **Persistent Settings:** Remember theme preference and last-active tab using localStorage.

Medium-Term (Version 3.0)

- **Backend Data Layer:** Node.js or Python Flask backend to serve real-time analytics data.
- **Multi-Platform Comparison:** Analyze competitors (Blinkit, Swiggy Instamart, BigBasket).
- **Predictive Analytics:** TensorFlow.js ML tab for traffic prediction using linear regression.
- **AI-Powered Recommendations:** Replace the static Recommendations tab with a live Claude API engine.

14. Project Summary

The following table provides a consolidated summary of all key metrics, technologies, and deliverables produced during this project.

Dashboard KPIs — Zepto.com Analysis

Metric Category	Metric Name	Value	Benchmark / Status
Traffic	Monthly Visitors	18.4 Million	Growing MoM
Traffic	Avg Session Duration	4 minutes 32 seconds	Above 3 min — Good
Traffic	Bounce Rate	38.2%	Below 40% — Good
Traffic	Pages per Session	6.8	Above 5 — Excellent
Traffic	Direct Traffic Share	68%	Strong brand recall
SEO	Domain Authority	72 / 100	Strong (target 80+)
SEO	Page Authority	64 / 100	Good
SEO	Total Backlinks	48,200	Healthy
SEO	Mobile Page Speed	62 / 100	Poor — needs fix
SEO	Top Keyword Rank	#1 for "zepto delivery"	Owned
Advertising	Click-Through Rate (CTR)	4.82%	Excellent (2x industry avg)
Advertising	Cost Per Click (CPC)	Rs 12.40	Efficient
Advertising	Customer Acquisition Cost	Rs 186	Acceptable
Advertising	Return on Ad Spend (ROAS)	3.8x	Profitable (target 4x)
Advertising	Return on Investment (ROI)	280%	Strong
Advertising	Conversion Rate (CVR)	12.17%	Above 10% — Good

Project Technical Summary

Aspect	Detail
File Type	Single HTML file — zepto_analytics_dashboard_v2.html
Technologies	HTML5, CSS3, JavaScript ES6+, Chart.js 4.4.1, Google Fonts
Charts	3 Chart.js charts: Traffic Trend (line), Campaign (bar), ROI Curve (line)
Interactivity	5-tab navigation + theme toggle + 5 live ROI sliders

Aspect	Detail
Theming	Full dark and light mode via CSS custom properties (14 token pairs)
Responsive	CSS Grid auto-fit — works on desktop, tablet, and mobile
Deployment	Open .html in browser — no server, no npm, no build step needed
GitHub	github.com/MeRajaKumar/Zepto_Traffic_Insights_Web_Analytics
Course	Web Analytics Lab — 24CAP-752 — MCA 4th Semester
Institution	University Institute of Computing, Chandigarh University

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16. Conclusion

The Zepto Web Analytics Dashboard successfully demonstrates the integration of web analytics theory with practical front-end development skills. Starting from a blank HTML file, the project evolved into a professional-grade, interactive analytics application with five fully populated sections, three dynamic charts, a live ROI simulator, and a complete dark/light theming system.

The analytical findings show that Zepto.com maintains strong direct traffic at 68%, a healthy domain authority of 72/100, and advertising campaigns that exceed industry benchmarks with a combined CTR of 4.82%, ROAS of 3.8x, and ROI of 280%. The most critical opportunities are mobile page speed optimization (currently 62%) and cart abandonment recovery (48% drop-off at checkout).

From a technical standpoint, the project proves that a production-quality, fully responsive, themeable analytics dashboard can be built using only HTML, CSS, and JavaScript — with zero build tools, no frameworks, and a single file. The codebase serves as a strong foundation for the future enhancements outlined in Section 13.

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Institute	University Institute of Computing, Chandigarh University, Gharuan, Mohali